CIS6008 - Analytics and Business Intelligence

Your Name / Student ID

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Contents

1.Introduction

here we analyse the given aviation dataset to

2.Task A

```
## 'data.frame':    200 obs. of  7 variables:
##  $ airport_traffic : num  224836 193087 232384 276151 188292 ...
##  $ avg_income      : num  69293 71729 77997 77646 48468 ...
##  $ fuel_price      : num  0.661 0.81 0.901 0.907 0.832 ...
##  $ avg_ticket_fare : num  280 213 285 304 267 ...
##  $ flight_frequency: num  139 110 122 111 111 ...
##  $ route_distance  : num  2060 1870 1524 1241 1779 ...
##  $ passenger_demand: num  124866 85800 126878 164130 97608 ...
```

as mentioned in the data dictionary this is Air transport dataset with 200 observations with 7 variables with all of them are numerical variables and passenger_demand being target(dependent) variable. and other variables are independent variables

```
##  airport_traffic    avg_income       fuel_price      avg_ticket_fare
##  Min.   : 69013   Min.   : 26105   Min.   :0.5293   Min.   :142.1
##  1st Qu.:164744   1st Qu.: 57730   1st Qu.:0.7809   1st Qu.:221.7
##  Median :199790   Median : 65946   Median :0.8884   Median :250.9
##  Mean   :197962   Mean   : 66030   Mean   :0.8872   Mean   :250.4
##  3rd Qu.:225043   3rd Qu.: 73247   3rd Qu.:0.9858   3rd Qu.:277.4
##  Max.   :336008   Max.   :111233   Max.   :1.3618   Max.   :355.3
##  flight_frequency route_distance   passenger_demand
##  Min.   : 71.52   Min.   : 341.5   Min.   :-60240
##  1st Qu.:109.38   1st Qu.:1298.3   1st Qu.: 68592
##  Median :122.61   Median :1579.1   Median :109586
##  Mean   :122.57   Mean   :1553.5   Mean   :107811
##  3rd Qu.:134.37   3rd Qu.:1849.4   3rd Qu.:140320
##  Max.   :170.54   Max.   :2475.9   Max.   :268128
```

Based on the summary statistic of the Air transport dataset, airport traffic range from a minimum of 69013 to a maximum of 336008, with a median of 199790 and a mean of 197962 The first and third quantiles are 164744and 225043 respectively. This distribution around the mean suggests a diverse airport traffic

The Average Income(USD per year) ranges from 26105 to 111233, with a median of 65946 and a mean of 66030. The first and third quantiles are 57730 and 73247, respectively.

Fuel Price(USD per liter) varies from 0.5293 to 1.3618, with a median of 0.8884 and a mean of 0.8872. The first and third quartiles are 0.7809 and 0.9858 respectively.

Average Ticket Fare spans from 142.1 to 355.3, with a median of 250.9 and a mean of 250.4.The first and third quartiles are 221.7 and 277.4, respectively.

Flight Frequency values range from 71.52 to 170.54, with a median of 122.61 and a mean of 122.57. The first and third quartiles are 109.38 and 134.37 respectively.

route_distance range from 341.5 to 2475.9 , with a median of 1579.1 and a mean of 1553.5. The first and third quartiles are 1298.3 and 1849.4 respectively.

Finally our target variable, Total Passenger Demand ranges from -60240(which is impossible in real life, because counts cannot be negative) to 268128 with a median of 109586 and a mean of 107811. The first and third quartiles are 68592 and 140320 respectively.

```
##    airport_traffic avg_income fuel_price avg_ticket_fare flight_frequency
## 1         224835.7   69293.45  0.6608359        280.2795         138.7657
## 2         193086.8   71729.41  0.8100937        213.1134         109.6791
## 3         232384.4   77996.61  0.9007866        284.7842         121.9224
## 4         276151.5   77645.62  0.9070471        304.2255         110.7545
## 5         188292.3   48467.97  0.8324902        266.5374         111.3101
## 6         188293.2   53746.10  0.9934275        325.0718         113.8166
## 7         278960.6   71180.42  0.7398569        219.0484         124.4427
## 8         238371.7   71165.43  0.8786431        200.2138         110.4250
## 9         176526.3   71180.57  0.9180443        178.8512         145.1151
## 10        227128.0  111232.78  0.9771658        309.8418         102.1079
##    route_distance passenger_demand
## 1        2059.742        124866.27
## 2        1869.853         85799.76
## 3        1523.852        126877.58
## 4        1241.225        164130.48
## 5        1779.289         97607.78
## 6        1657.394         91178.41
## 7        1858.077        117150.75
## 8        1754.069         99517.98
## 9        1919.821         43882.48
## 10       1285.906        200988.47
```

1.Univariate analysis

we will analyze each variable graphically and get initial insights from graphs then we develop hypotheses about variable distributions using graphical insights and then we will confirm them using Hypothesis testing statistics, specially normality test and other distribution tests.

basic hypothesis testing,

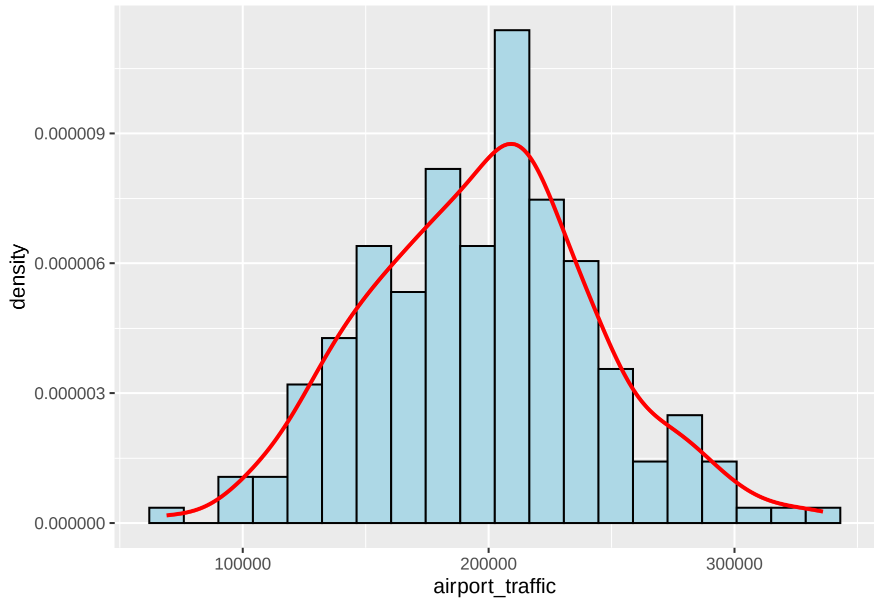
H0: variable follows a certain distribution

H1: variable does not follow a certain distribution

note that we will consider significance level(alpha) 5% in these tests

if the test statistic's p-value is less than 5% we reject our null hypothesis(H0) at 5% significance level, favoring our alternative hypothesis(H1) or in other words if we can't be at least 95% sure that variable follows that certain distribution we will reject null hypothesis. if the test statistic's p-value is greater than or equal to 5% we reject our alternative hypothesis(H1) at 5% significance level favoring our null hypothesis(H0) or in other words, if we can be 95% sure that variable follows that certain distribution we will accept our null hypothesis.

1.airport_traffic Variable



from looking at the histogram it seems like the airport traffic is normally distributed.but we need to clarify the above claim using normality tests.

hypothesis testing,

H0:airport traffic variable follows a normal distribution

H1:airport traffic variable does follows a normal distribution

a)Shapiro-Wilk Test

```
##
##  Shapiro-Wilk normality test
##
## data:  df$airport_traffic
## W = 0.99556, p-value = 0.829
```

since p-value = 0.829> 0.05(alpha) we fail to reject our null hypothesis. that is shapiro wilk test strongly suggests that our airport traffic variable is normally distributed.

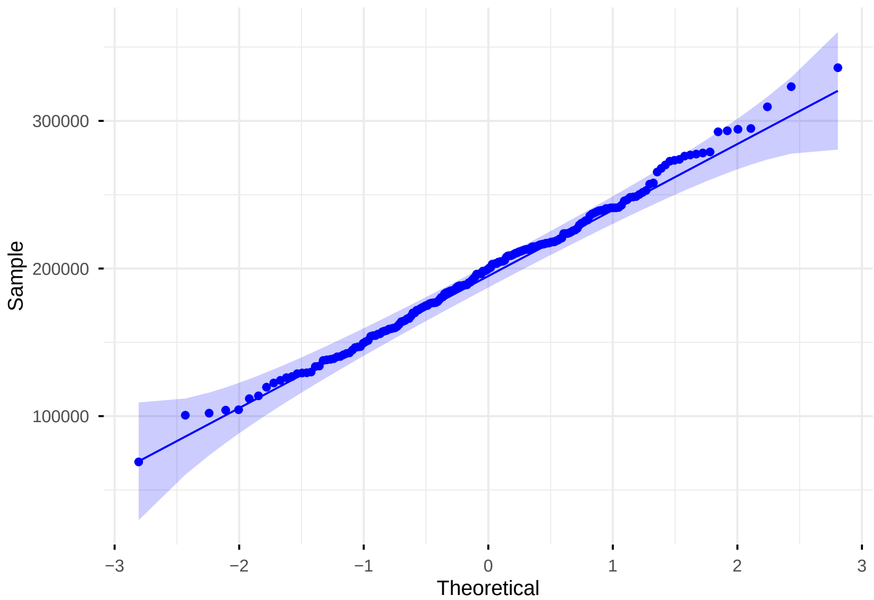
b)Anderson-Darling Test

```
##
##  Anderson-Darling normality test
##
## data:  df$airport_traffic
## A = 0.2747, p-value = 0.6586
```

since p-value = 0.6586 > 0.05(alpha) we fail to reject our null hypothesis. that is anderson test also suggests that our airport traffic variable is normally distributed.

since both tests suggest that airport traffic is normally distributed we can verify it one more time using a QQ plot where we plot theoretical normal quantiles against our data sample quantiles

c)Q-Q plot

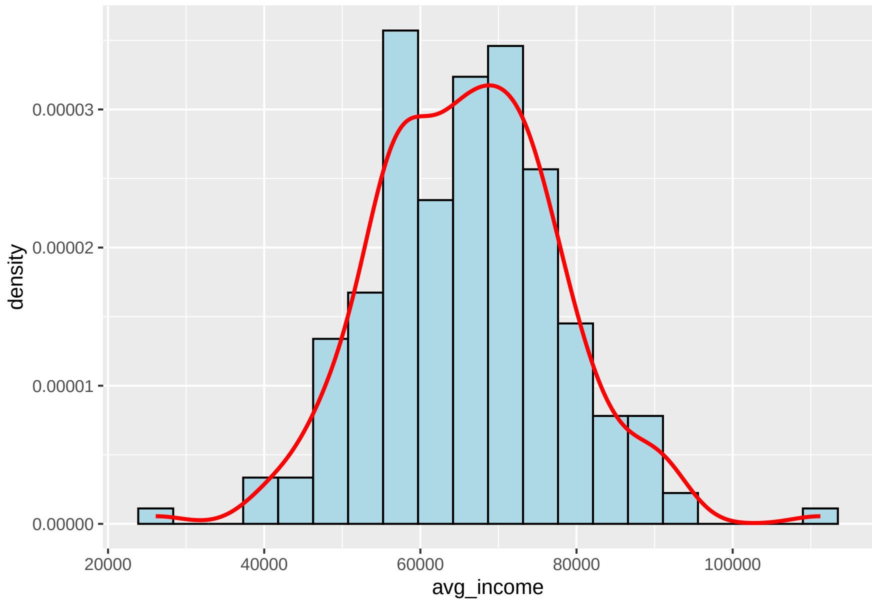


from the Q-Q we can again conclude that our airport traffic variable is normally distributed as it almost perfectly align with the theoretical normal line shown in the Q-Q plot.

```
##  airport_traffic   avg_income       fuel_price      avg_ticket_fare
##  Min.   : 69013   Min.   : 26105   Min.   :0.5293   Min.   :142.1
##  1st Qu.:164744   1st Qu.: 57730   1st Qu.:0.7809   1st Qu.:221.7
##  Median :199790   Median : 65946   Median :0.8884   Median :250.9
##  Mean   :197962   Mean   : 66030   Mean   :0.8872   Mean   :250.4
##  3rd Qu.:225043   3rd Qu.: 73247   3rd Qu.:0.9858   3rd Qu.:277.4
##  Max.   :336008   Max.   :111233   Max.   :1.3618   Max.   :355.3
##  flight_frequency route_distance   passenger_demand
```

```
##  Min.   : 71.52   Min.   : 341.5   Min.   :-60240
##  1st Qu.:109.38   1st Qu.:1298.3   1st Qu.: 68592
##  Median :122.61   Median :1579.1   Median :109586
##  Mean   :122.57   Mean   :1553.5   Mean   :107811
##  3rd Qu.:134.37   3rd Qu.:1849.4   3rd Qu.:140320
##  Max.   :170.54   Max.   :2475.9   Max.   :268128
```

2.avg_income Variable



from looking at the histogram it seems like the avg_income Variable is normally distributed as it shows a bell curved shape but we need to clarify this claim using several normality tests and Q-Q plot.

hypothesis testing,

H0: avg_income variable follows a normal distribution

H1: avg_income variable does follows a normal distribution

a)Shapiro-Wilk Test

```
##
##  Shapiro-Wilk normality test
##
## data:  df$avg_income
## W = 0.99101, p-value = 0.2498
```

since p-value = 0.2498 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is shapiro wilk test suggests that our avg_income variable is normally distributed.

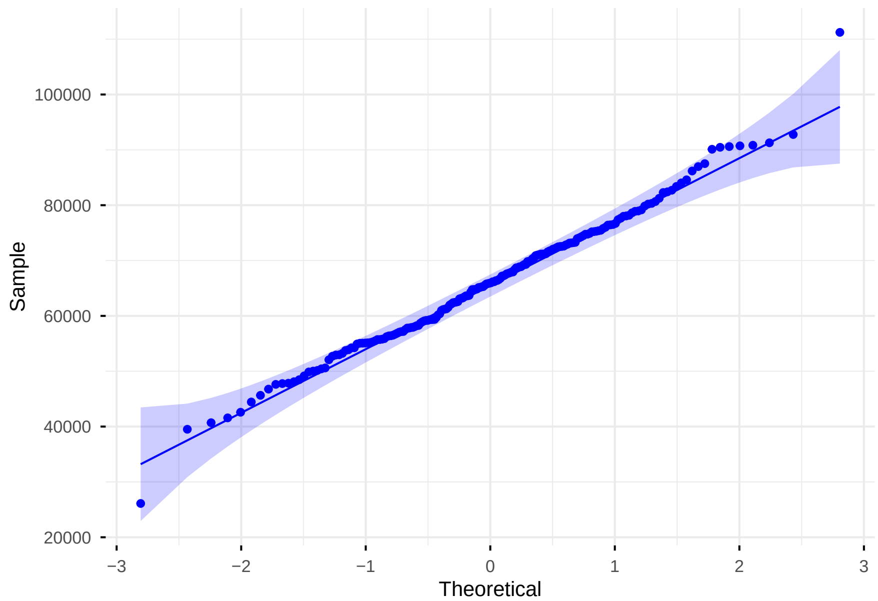
b)Anderson-Darling Test

```
##
##  Anderson-Darling normality test
##
## data:  df$avg_income
## A = 0.32237, p-value = 0.5256
```

since p-value = 0.5256 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is Anderson-Darling Test also suggests that our avg_income variable is normally distributed.

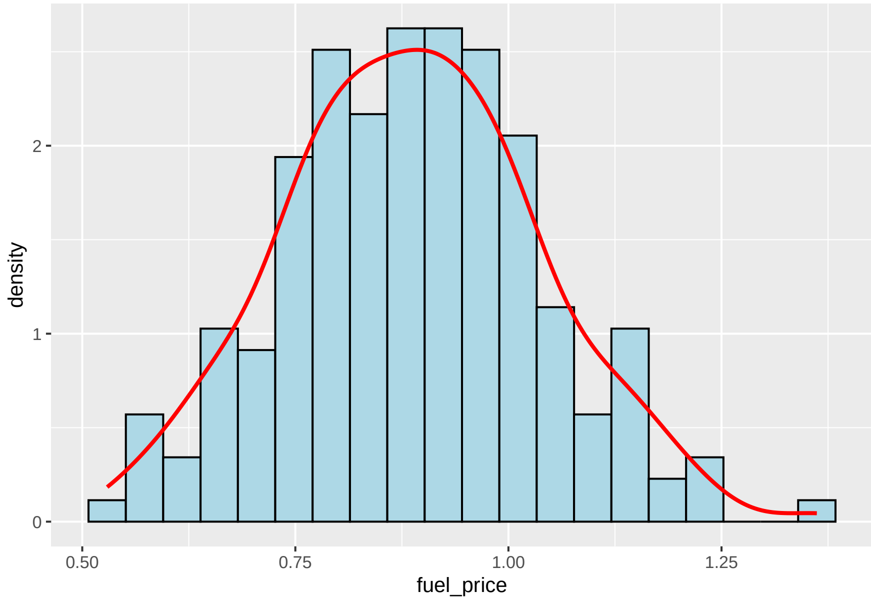
since both tests suggest that average income is normally distributed we can verify this claim one more time using a Q-Q plot.

c)Q-Q plot



from the Q-Q we can see that our avg_income variable is normally distributed as it perfectly aligns with the theoretical normal quantiles as shown in the Q-Q plot. so we can conclude that average income is also normally distributed.

3.fuel_price Variable



from looking at the histogram it seems like the fuel_price Variable is also normally distributed as the bell shape is present in the histogram but we need to clarify this claim using several normality tests and Q-Q plot.

hypothesis testing,

H0: fuel_price variable follows a normal distribution

H1: fuel_price variable does follows a normal distribution

a)Shapiro-Wilk Test

```
##
##  Shapiro-Wilk normality test
##
## data:  df$fuel_price
## W = 0.99638, p-value = 0.9218
```

since p-value = 0.9218 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is shapiro-wilk test statistic strongly suggests that fuel price variable is normally distributed.

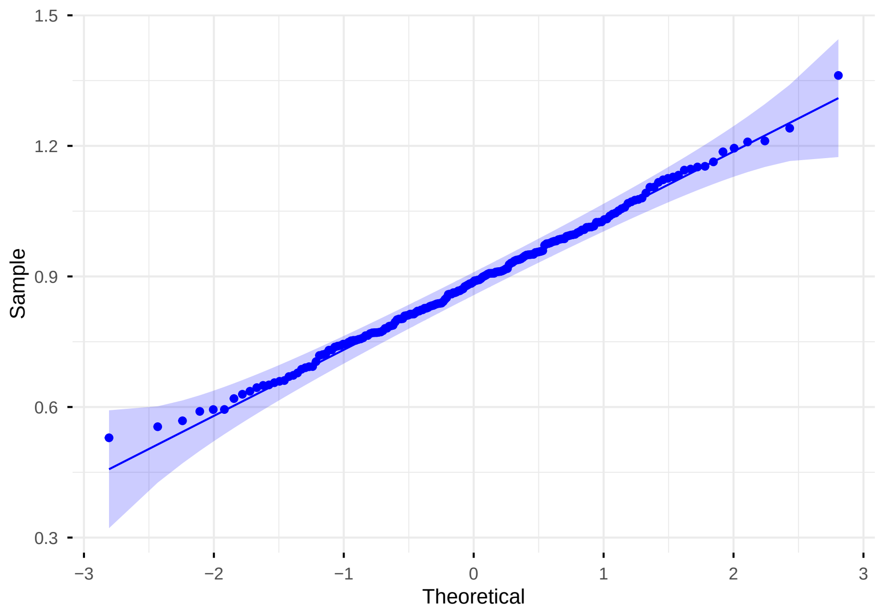
b)Anderson-Darling Test

```
##
##  Anderson-Darling normality test
##
```

```
## data:  df$fuel_price
## A = 0.13451, p-value = 0.9789
```

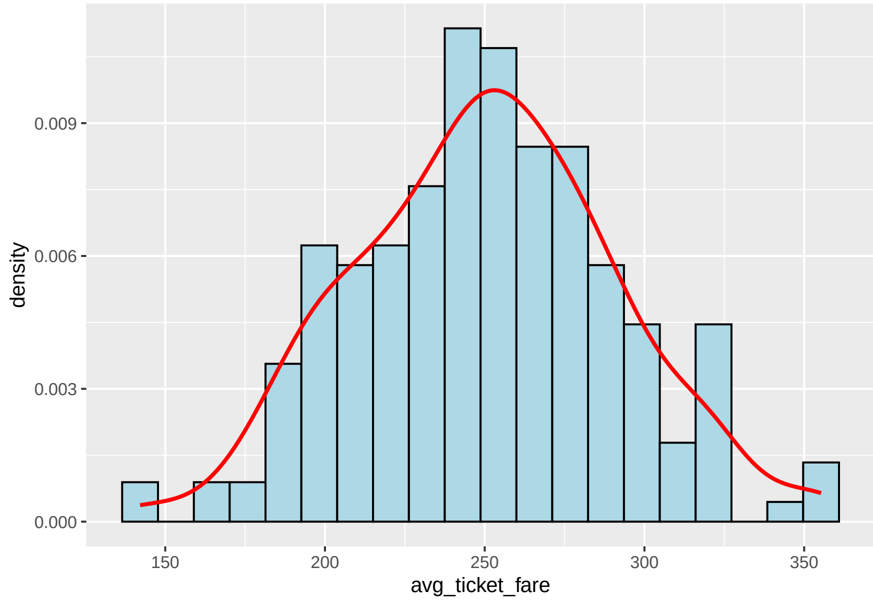
since p-value = 0.9789 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is Anderson-Darling test statistic also suggests that fuel price variable is normally distributed. since both tests suggest that fuel price is normally distributed we can verify this claim one more time using a Q-Q plot.

c)Q-Q plot



from the Q-Q we can see that our fuel_price variable is normally distributed as it perfectly align with the theoretical normal line, this suggests our fuel price variable is normally distributed.

4.avg_ticket_fare Variable



from looking at the histogram it seems like the avg_ticket_fare Variable is also normally distributed as the bell shape is present in the histogram but we need to clarify this claim using several normality tests and Q-Q plot.

hypothesis testing,

H0: avg_ticket_fare variable follows a normal distribution

H1: avg_ticket_fare variable does follows a normal distribution

a)Shapiro-Wilk Test

```
##
##  Shapiro-Wilk normality test
##
## data:  df$avg_ticket_fare
## W = 0.99519, p-value = 0.7774
```

since p-value = 0.7774 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is shapiro-wilk test strongly suggests that our avg_ticket_fare variable is normally distributed.

b)Anderson-Darling Test

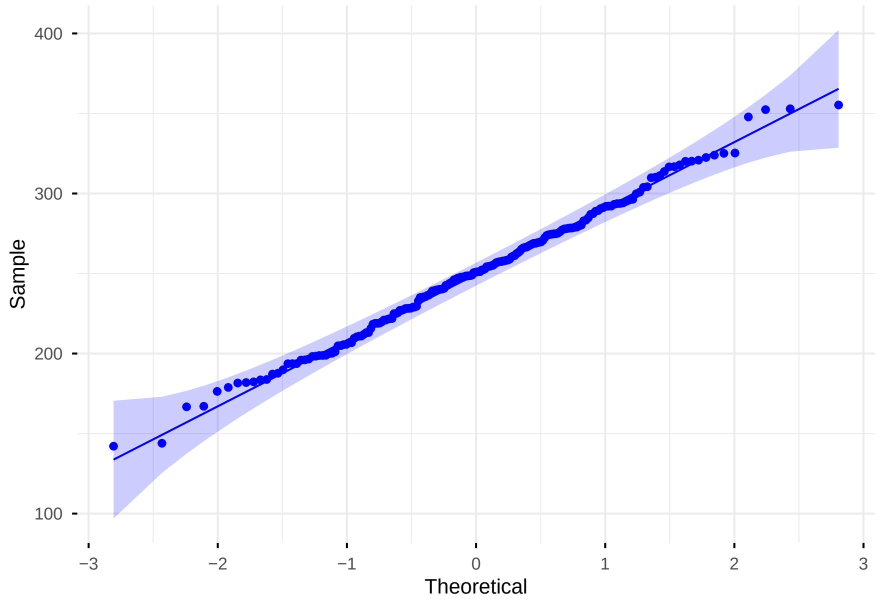
```
##
##  Anderson-Darling normality test
##
```

```
## data:  df$avg_ticket_fare
## A = 0.20623, p-value = 0.8683
```

since p-value = 0.8683 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is Anderson-Darling Test also suggests that our avg_ticket_fare variable is normally distributed.

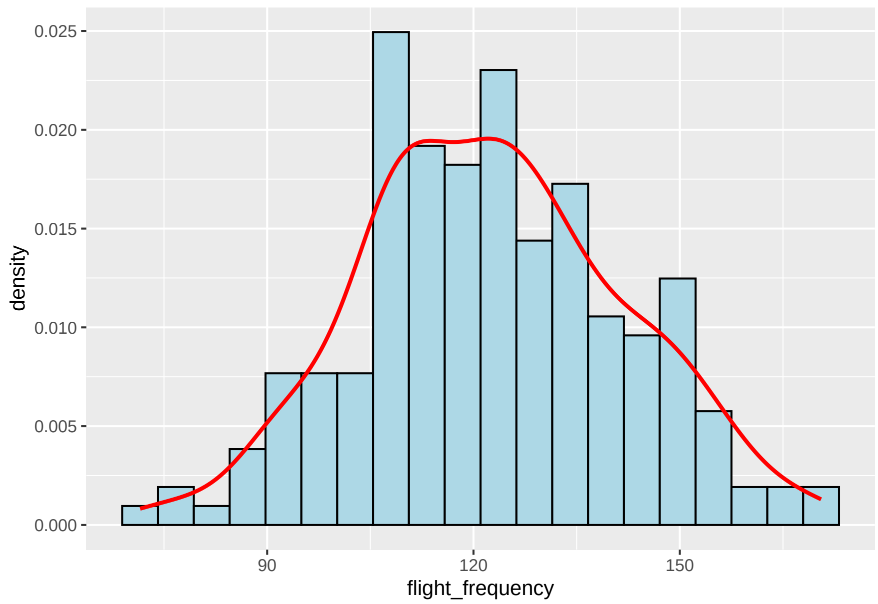
since both tests suggest that avg_ticket_fare is normally distributed we can verify this claim one more time using a Q-Q plot

c)Q-Q plot



from the Q-Q we can see that our avg_ticket_fare variable is normally distributed as it mostly align with the theoretical normal line and there almost no deviances from theoretical normal line as shown in the Q-Q plot. and also most ofdata points are inside the confidence interval region, this suggest our avg_ticket_fare variable is normally distributed.

5.flight_frequency Variable



from looking at the histogram it seems like the flight_frequency Variable is normally distributed as the bell shape is present in histogram but we need to clarify this claim using several normality tests and Q-Q plot.

hypothesis testing,

H0: flight_frequency variable follows a normal distribution

H1: flight_frequency variable does follows a normal distribution

a)Shapiro-Wilk Test

```
##
##  Shapiro-Wilk normality test
##
## data:  df$flight_frequency
## W = 0.994, p-value = 0.6003
```

since p-value = 0.6003 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is shapiro-wilk test suggests that our flight_frequency variable is normally distributed.

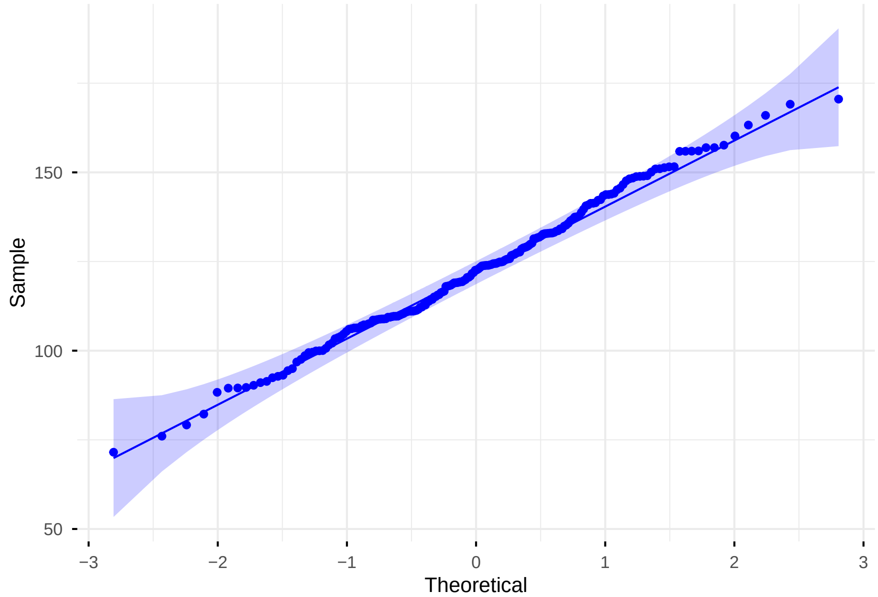
b)Anderson-Darling Test

```
##
##  Anderson-Darling normality test
##
## data:  df$flight_frequency
## A = 0.41248, p-value = 0.336
```

since p-value = 0.336 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is Anderson-Darling test also suggests that our flight_frequency variable is normally distributed.

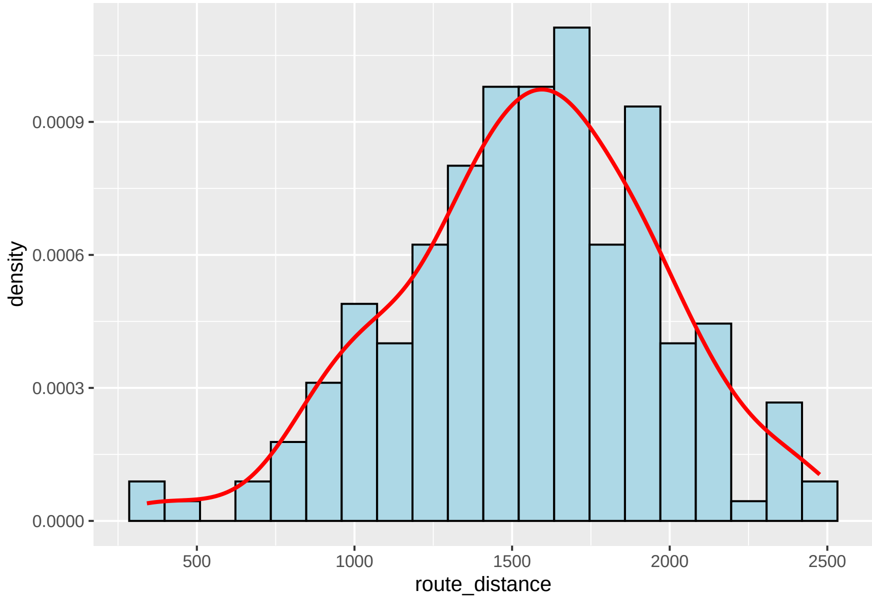
since both tests suggest that flight_frequency is normally distributed we can verify this claim one more time using a graphical method which is Q-Q plot.

c)Q-Q plot



from the Q-Q we can see that our flight_frequency variable is normally distributed as it mostly align with the theoretical normal line and there almost no deviances from theoretical normal line as shown in the Q-Q plot. and also most of data points are inside the confidence interval region, this suggest our flight_frequency variable is normally distributed.

6.route_distance



from looking at the histogram it seems like the route_distance Variable is normally distributed as the bell shape is present in histogram but we need to clarify this claim using several normality tests and Q-Q plot.

hypothesis testing,

H0: route_distance variable follows a normal distribution

H1: route_distance variable does follows a normal distribution

a)Shapiro-Wilk Test

```
##
##  Shapiro-Wilk normality test
##
## data:  df$route_distance
## W = 0.99284, p-value = 0.4398
```

since p-value = 0.4398 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is shapiro-wilk test suggests that our route_distance variable is normally distributed.

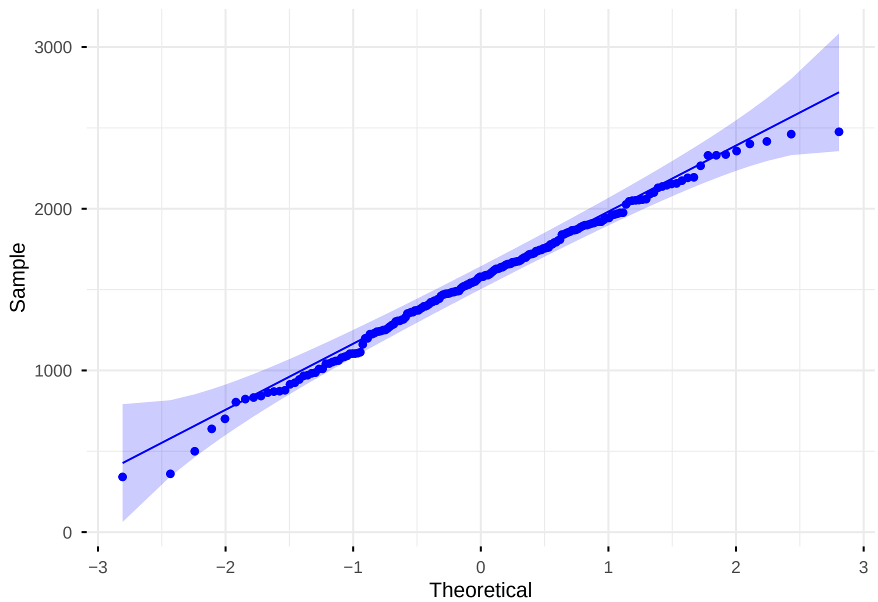
b)Anderson-Darling Test

```
##
##  Anderson-Darling normality test
##
## data:  df$route_distance
## A = 0.27734, p-value = 0.6498
```

since p-value = 0.6498 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is Anderson-Darling test also suggests that our route_distance variable is normally distributed.

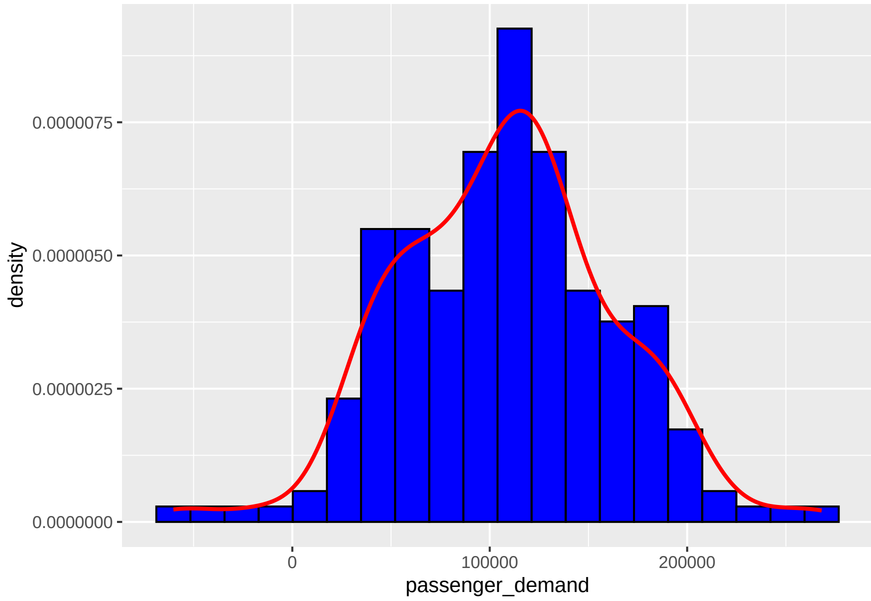
since both tests suggest that route_distance is normally distributed we can verify this claim one more time using a graphical method which is Q-Q plot.

c)Q-Q plot



from the Q-Q we can see that our route_distance variable is normally distributed as it mostly align with the theoretical normal line and there almost no deviances from theoretical normal line as shown in the Q-Q plot. and also most ofdata points are inside the confidence interval region, this suggest our route_distance variable is normally distributed.

7.passenger_demand(Target) Variable



from looking at the histogram it seems like the passenger_demand(Target) Variable is normally distributed as follows the symmetric bell shape which can be seen in the histogram. but we need to clarify this claim using several normality tests and Q-Q plot.

hypothesis testing,

H0: passenger_demand variable follows a normal distribution

H1: passenger_demand variable does follows a normal distribution

a)Shapiro-Wilk Test

```
##
##  Shapiro-Wilk normality test
##
## data:  df$passenger_demand
## W = 0.99272, p-value = 0.4249
```

since p-value = 0.4249 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is shapiro-wilk test suggests that our passenger_demand variable is normally distributed.

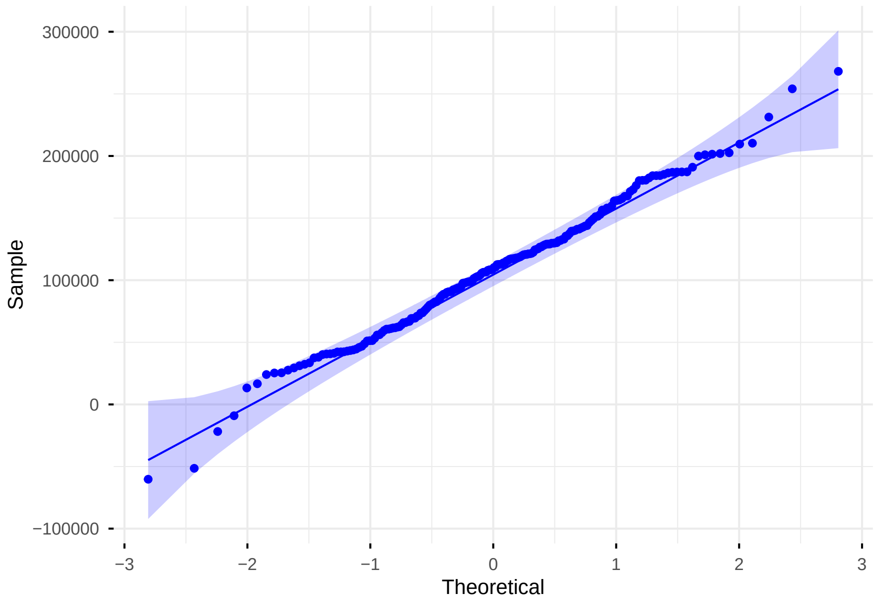
b)Anderson-Darling Test

```
##
##  Anderson-Darling normality test
##
## data:  df$passenger_demand
## A = 0.38674, p-value = 0.3862
```

since p-value = 0.3862 > 0.05(alpha) we fail to reject our null hypothesis(H0). that is Anderson-Darling test also suggests that our passenger_demand variable is normally distributed.

since both tests suggest that passenger_demand is normally distributed we can verify this claim one more time using a Q-Q plot.

c)Q-Q plot



from the Q-Q we can see that our passenger_demand variable is normally distributed as it mostly align with the theoretical normal line and there almost no deviances from theoretical normal line as shown in the Q-Q plot. and also most of data points are inside the confidence interval region, this suggest our passenger_demand(Target) variable is normally distributed.

2.Bivariate Analysis(Correlation analysis)

in this section we will analyse the relationship between our target variable(Total Passenger Demand) and each independent variable using graphical methods and some correlation tests.

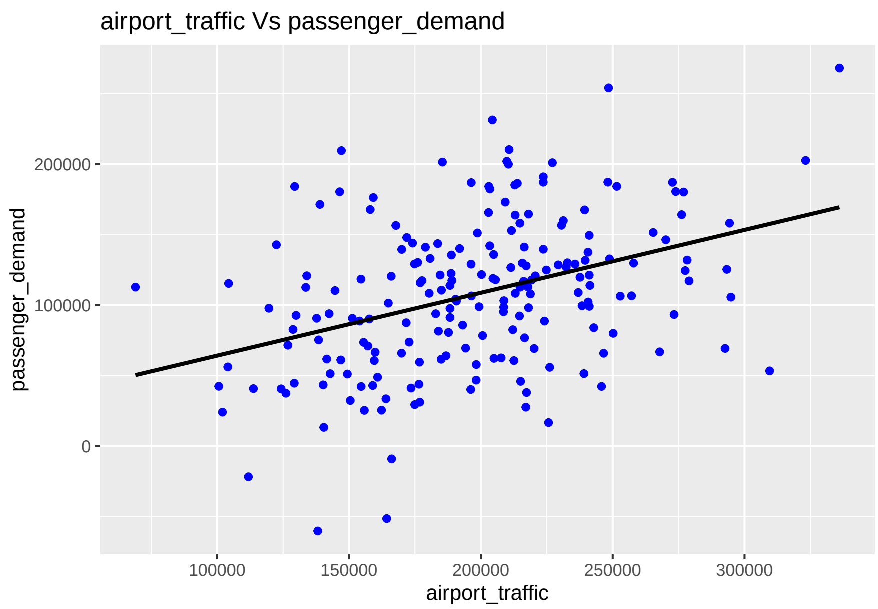
here we will first inspect what kind of association is there between all independent variables and dependent variable using scatter plots. then we will use pearson correlation test as it shows linear association between two variables. but from the scatter plots if there seems to be a non linear relationship we will use kendal's tau test and spearman rho test statistics to further measeure the association.

note that all test statistic scores are in between -1 and 1 and, 1 suggests strong positive correlation 0 suggests no correlation at all(2 variables are independent) -1 suggests strong negative correlation

and magnitude of score suggests how strong is the relationship and sign suggests the relationship negative or positive(ex: +0.9 means a strong positive association, -0.9 represents a strong negatvie association).

```
##  airport_traffic    avg_income       fuel_price      avg_ticket_fare
##  Min.    : 69013   Min.    : 26105   Min.    :0.5293   Min.    :142.1
##  1st Qu.:164744   1st Qu.: 57730   1st Qu.:0.7809   1st Qu.:221.7
##  Median :199790   Median : 65946   Median :0.8884   Median :250.9
##  Mean    :197962   Mean    : 66030   Mean    :0.8872   Mean    :250.4
##  3rd Qu.:225043   3rd Qu.: 73247   3rd Qu.:0.9858   3rd Qu.:277.4
##  Max.    :336008   Max.    :111233   Max.    :1.3618   Max.    :355.3
##  flight_frequency route_distance   passenger_demand
##  Min.    : 71.52   Min.    : 341.5   Min.    :-60240
##  1st Qu.:109.38   1st Qu.:1298.3   1st Qu.: 68592
##  Median :122.61   Median :1579.1   Median :109586
##  Mean    :122.57   Mean    :1553.5   Mean    :107811
##  3rd Qu.:134.37   3rd Qu.:1849.4   3rd Qu.:140320
##  Max.    :170.54   Max.    :2475.9   Max.    :268128
```

1.airport_traffic Vs passenger_demand



from the scatter plot we can see that airport_traffic and passenger_demand have a linear a positive relationship we can clarify this using a above mentioned test

hypothesis testing,

H0: airport_traffic and passenger_demand are independent

H1: airport_traffic and passenger_demand are not independent

a)Pearson test

```
##
##  Pearson's product-moment correlation
##
## data:  df$airport_traffic and df$passenger_demand
## t = 5.8771, df = 198, p-value = 0.00000001746
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2605941 0.4975381
## sample estimates:
##      cor
## 0.385401
```

r = 0.385401 suggests that airport_traffic and passenger_demand has a somewhat positive association and it is significant at 5% significant level

b)Kendall's tau test

```
##
##  Kendall's rank correlation tau
##
## data:  df$airport_traffic and df$passenger_demand
## z = 5.2964, p-value = 0.0000001181
## alternative hypothesis: true tau is not equal to 0
## sample estimates:
##       tau
## 0.2518593
```

tau = 0.2518593 suggests that airport_traffic and passenger_demand has a somewhat positive association and it is significant at 5% significant level

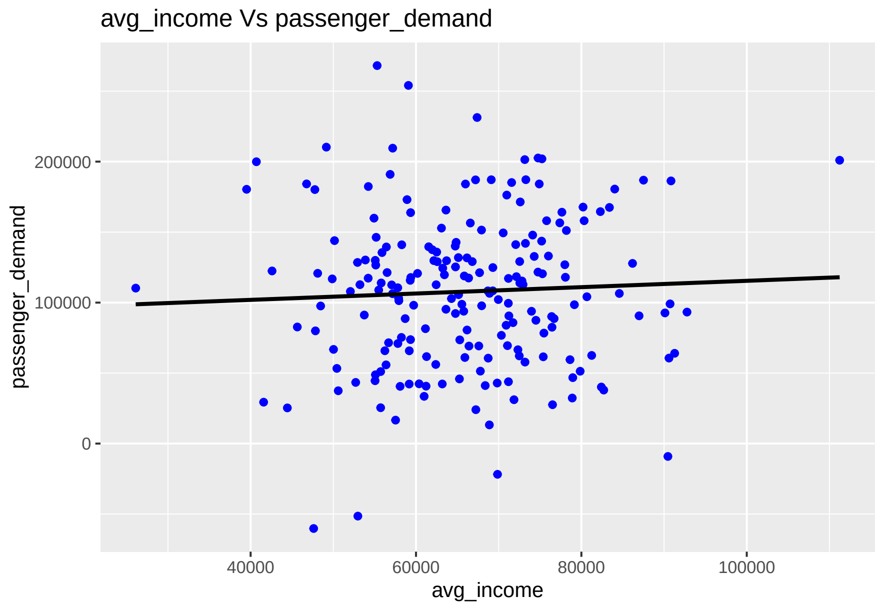
c)Spearman test

```
##
##  Spearman's rank correlation rho
##
## data:  df$airport_traffic and df$passenger_demand
## S = 836036, p-value = 0.00000006781
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.3729573
```

rho = 0.3729573 suggests that airport_traffic and passenger_demand has a somewhat positive association and it is significant at 5% significant level

since all tests suggest that airport_traffic and passenger_demand are not independent(reject H0 because all p values < 0.05) and they have a weak positive relationship. we can say that the relationship is mostly linear by looking at the scatter plot.

2.avg_income Vs passenger_demand



from the scatter plot we can see that avg_income and passenger_demand do not have an association at all because the relationship line is almost a flat line

hypothesis testing,

H0: avg_income and passenger_demand are independent

H1: avg_income and passenger_demand are not independent

a)Pearson test

```
##
##  Pearson's product-moment correlation
##
## data:  df$avg_income and df$passenger_demand
## t = 0.69808, df = 198, p-value = 0.4859
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.08980908  0.18700478
## sample estimates:
```

```
##        cor
## 0.04954933
```

r = 0.04954933 suggests that avg_income and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p = 0.4859>0.05)

b)Kendall's tau test

```
##
##  Kendall's rank correlation tau
##
## data:  df$avg_income and df$passenger_demand
## z = 0.67843, p-value = 0.4975
## alternative hypothesis: true tau is not equal to 0
## sample estimates:
##        tau
## 0.03226131
```

tau = 0.03226131 suggests that avg_income and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.4975 > 0.05)

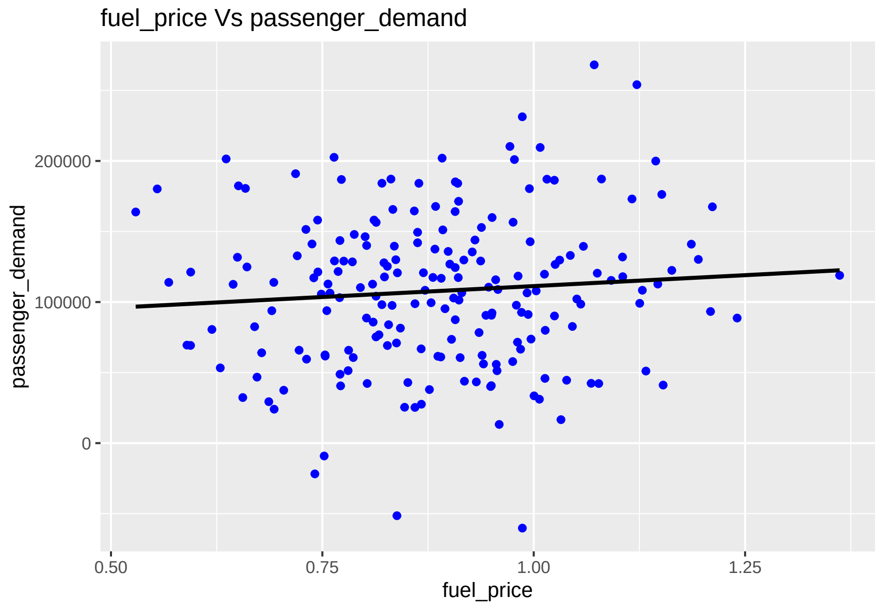
c)Spearman test

```
##
##  Spearman's rank correlation rho
##
## data:  df$avg_income and df$passenger_demand
## S = 1275310, p-value = 0.5405
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## 0.04349359
```

rho = 0.04349359 suggests that avg_income and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.5405 > 0.05)

since all tests suggest that avg_income and passenger_demand are independent(fail to reject H0 because all p values > 0.05) we can conclude that there is no association between avg_income and passenger_demand.

3.fuel_price Vs passenger_demand



from the scatter plot we can see that fuel_price and passenger_demand do not have an association at all because the relationship line is almost a flat line

hypothesis testing,

H0: fuel_price and passenger_demand are independent

H1: fuel_price and passenger_demand are not independent

a)Pearson test

```
##
##  Pearson's product-moment correlation
##
## data:  df$fuel_price and df$passenger_demand
## t = 1.21, df = 198, p-value = 0.2277
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.05370386  0.22178029
## sample estimates:
##        cor
## 0.08567551
```

r = 0.08567551 suggests that fuel_price and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level($p = 0.2277 > 0.05$)

b)Kendall's tau test

```
##
##  Kendall's rank correlation tau
##
## data:  df$fuel_price and df$passenger_demand
## z = 0.82848, p-value = 0.4074
## alternative hypothesis: true tau is not equal to 0
## sample estimates:
##        tau
## 0.03939698
```

tau = 0.03939698 suggests that fuel_price and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.4074 > 0.05)

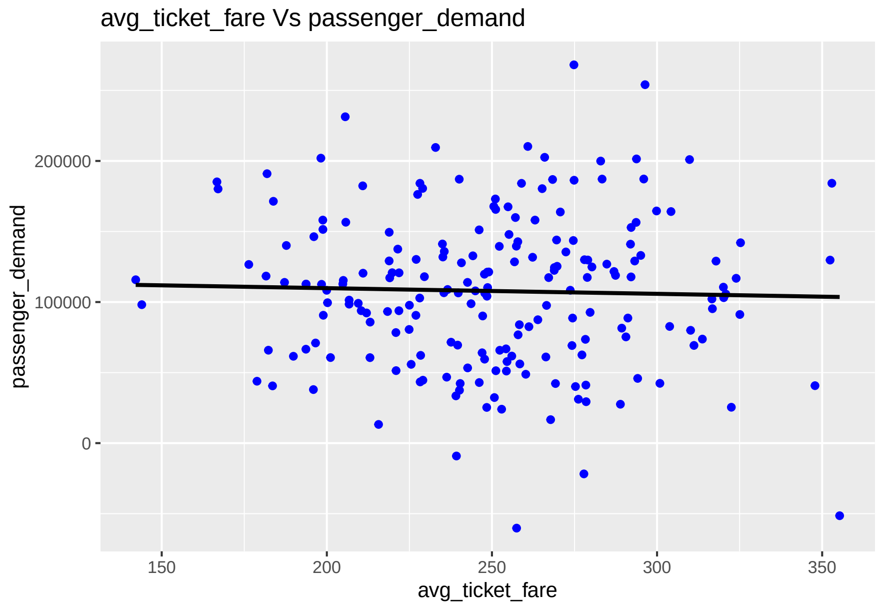
c)Spearman test

```
##
##  Spearman's rank correlation rho
##
## data:  df$fuel_price and df$passenger_demand
## S = 1253414, p-value = 0.399
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.059916
```

rho =0.059916 suggests that fuel_price and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.399 > 0.05)

since all tests suggest that fuel_price and passenger_demand are independent(fail to reject H0 because all p values > 0.05) we can conclude that there is no association between fuel_price and passenger_demand.

4.avg_ticket_fare Vs passenger_demand



from the scatter plot we can see that avg_ticket_fare and passenger_demand do not have an association at all because the relationship line is almost a flat line

hypothesis testing,

H0: avg_ticket_fare and passenger_demand are independent

H1: avg_ticket_fare and passenger_demand are not independent

a)Pearson test

```
##
##  Pearson's product-moment correlation
##
## data:  df$avg_ticket_fare and df$passenger_demand
## t = -0.4279, df = 198, p-value = 0.6692
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.1684264  0.1088042
## sample estimates:
##         cor
## -0.03039567
```

r = -0.03039567 suggests that avg_ticket_fare and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p = 0.6692 > 0.05)

b)Kendall's tau test

```
##
##  Kendall's rank correlation tau
##
## data:  df$avg_ticket_fare and df$passenger_demand
## z = 0.15006, p-value = 0.8807
## alternative hypothesis: true tau is not equal to 0
## sample estimates:
##         tau
## 0.007135678
```

tau = 0.007135678 suggests that avg_ticket_fare and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.8807 > 0.05)

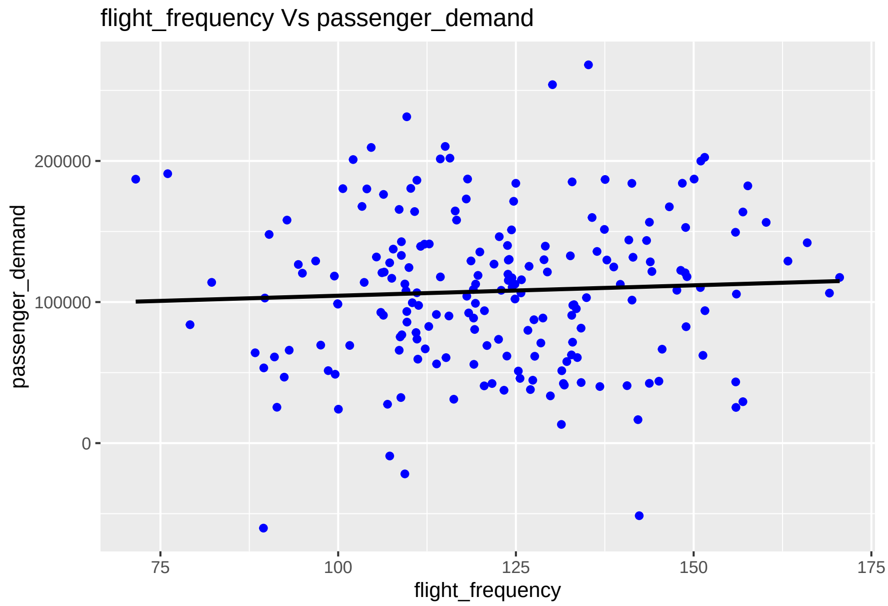
c)Spearman test

```
##
##  Spearman's rank correlation rho
##
## data:  df$avg_ticket_fare and df$passenger_demand
## S = 1315870, p-value = 0.8541
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## 0.01307283
```

rho =0.01307283 suggests that avg_ticket_fare and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.8541 > 0.05)

since all tests suggest that avg_ticket_fare and passenger_demand are independent(fail to reject H0 because all p values > 0.05) we can conclude that there is no association between avg_ticket_fare and passenger_demand.

5.flight_frequency Vs passenger_demand



from the scatter plot we can see that flight_frequency and passenger_demand do not have an association at all because the relationship line is almost a flat line

hypothesis testing,

H0: flight_frequency and passenger_demand are independent

H1: flight_frequency and passenger_demand are not independent

a)Pearson test

```
##
##  Pearson's product-moment correlation
##
## data:  df$flight_frequency and df$passenger_demand
## t = 0.7356, df = 198, p-value = 0.4628
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.08716655  0.18957374
## sample estimates:
##        cor
## 0.05220582
```

r = 0.05220582 suggests that flight_frequency and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p = 0.4628 > 0.05)

b)Kendall's tau test

```
##
##  Kendall's rank correlation tau
##
## data:  df$flight_frequency and df$passenger_demand
## z = 0.59389, p-value = 0.5526
## alternative hypothesis: true tau is not equal to 0
## sample estimates:
##        tau
## 0.02824121
```

tau = 0.02824121 suggests that flight_frequency and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.5526 > 0.05)

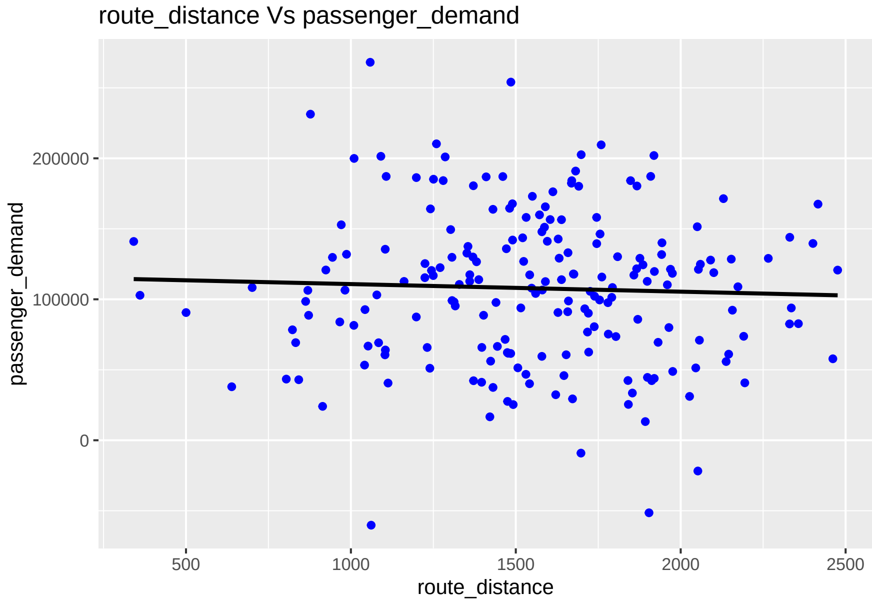
c)Spearman test

```
##
##  Spearman's rank correlation rho
##
## data:  df$flight_frequency and df$passenger_demand
## S = 1274890, p-value = 0.5376
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.0438086
```

rho =0.0438086 suggests that flight_frequency and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.5376 > 0.05)

since all tests suggest that flight_frequency and passenger_demand are independent(fail to reject H0 because all p values > 0.05) we can conclude that there is no association between flight_frequency and passenger_demand.

6.route_distance Vs passenger_demand



from the scatter plot we can see that route_distance and passenger_demand do not have an association at all because the relationship line is almost a flat line

hypothesis testing,

H0: route_distance and passenger_demand are independent

H1: route_distance and passenger_demand are not independent

a)Pearson test

```
##
##  Pearson's product-moment correlation
##
## data:  df$route_distance and df$passenger_demand
## t = -0.5797, df = 198, p-value = 0.5628
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.17888217  0.09813873
## sample estimates:
##         cor
## -0.04116275
```

r = -0.04116275 suggests that route_distance and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p = 0.5628 > 0.05)

b)Kendall's tau test

```
##
##  Kendall's rank correlation tau
##
## data:  df$route_distance and df$passenger_demand
## z = -0.41213, p-value = 0.6802
## alternative hypothesis: true tau is not equal to 0
## sample estimates:
##         tau
## -0.01959799
```

tau = -0.01959799 suggests that route_distance and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.6802 > 0.05)

c)Spearman test

```
##
##  Spearman's rank correlation rho
##
## data:  df$route_distance and df$passenger_demand
## S = 1370690, p-value = 0.6932
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.0280432
```

rho = -0.0280432 suggests that route_distance and passenger_demand are independent and it is not significant(i.e. no significant association) at 5% significant level(p-value = 0.6932 > 0.05)

since all tests suggest that route_distance and passenger_demand are independent(fail to reject H0 because all p values > 0.05) we can conclude that there is no association between route_distance and passenger_demand.

Now we will see how each variable is correlated with other using a correlation map

as we can see from the correlation no independent variable is highly correlated with other independent variables, so there won't be any multicolinearity issues.

3.Regression analysis

since each independent variables are in different units, to make it fair to a regression we have to standardize

```
##  airport_traffic      avg_income          fuel_price         avg_ticket_fare
##  Min.   :-2.77010   Min.   :-3.370945   Min.   :-2.400290   Min.   :-2.65390
##  1st Qu.:-0.71359   1st Qu.:-0.700814   1st Qu.:-0.712468   1st Qu.:-0.70170
##  Median : 0.03929   Median :-0.007121   Median : 0.008103   Median : 0.01304
##  Mean   : 0.00000   Mean   : 0.000000   Mean   : 0.000000   Mean   : 0.00000
##  3rd Qu.: 0.58176   3rd Qu.: 0.609281   3rd Qu.: 0.661686   3rd Qu.: 0.66323
##  Max.   : 2.96555   Max.   : 3.816463   Max.   : 3.183503   Max.   : 2.57305
##  flight_frequency    route_distance      passenger_demand
##  Min.   :-2.669779   Min.   :-2.94072   Min.   :-3.12037
##  1st Qu.:-0.689538   1st Qu.:-0.61899   1st Qu.:-0.72823
##  Median : 0.002391   Median : 0.06224   Median : 0.03295
##  Mean   : 0.000000   Mean   : 0.00000   Mean   : 0.00000
##  3rd Qu.: 0.617266   3rd Qu.: 0.71806   3rd Qu.: 0.60361
##  Max.   : 2.509256   Max.   : 2.23825   Max.   : 2.97675
```

from the correlation analysis we identified that there is no significant association between target variable and independent variables except airport_traffic. so there is no point of carrying out a single variable linear regression analysis for other variables except airport_traffic.

Only airport_traffic as a independent variable

```
##
## Call:
## lm(formula = passenger_demand ~ airport_traffic, data = df_standard)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.67812 -0.58806  0.00017  0.60106  2.31022
##
## Coefficients:
##                                Estimate              Std. Error t value
## (Intercept)     -0.00000000000000009993  0.06541278312926997185   0.000
## airport_traffic  0.38540095293429915690  0.06557693089835127387   5.877
##                     Pr(>|t|)
## (Intercept)                1
## airport_traffic 0.0000000175 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9251 on 198 degrees of freedom
## Multiple R-squared:  0.1485, Adjusted R-squared:  0.1442
## F-statistic: 34.54 on 1 and 198 DF,  p-value: 0.00000001746
```

```
## [1] 540.4138
```

since R-squared of the model is 0.1485 means airport traffic variable alone can explain 14.85% variability in the passenger demand data. adjusted R-squared is 0.1442 and AIC 540.4138

even though adding other variables to the above model systematically doesn't make much difference (because they are mostly independent with the target variable) we can add and see which model makes the most predictive accuracy and find out overall best model using AIC and R squared value

Here we will use forward selection method,

```
##
## Call:
## lm(formula = passenger_demand ~ airport_traffic + avg_income,
##     data = df_standard)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.66468 -0.59543 -0.01419  0.60314  2.31858
##
## Coefficients:
```

```
##                                 Estimate            Std. Error t value
## (Intercept)     -0.0000000000000001018  0.0655721484113572922   0.000
## airport_traffic  0.3841642958168862698  0.0660362853069394601   5.817
## avg_income       0.0129973754520843632  0.0660362853069395017   0.197
##                       Pr(>|t|)
## (Intercept)              1.000
## airport_traffic 0.0000000238 ***
## avg_income               0.844
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9273 on 197 degrees of freedom
## Multiple R-squared:  0.1487, Adjusted R-squared:  0.1401
## F-statistic: 17.21 on 2 and 197 DF,  p-value: 0.0000001297
```

```
## [1] 542.3745
```

adjusted R squared = 0.1401 decreased a bit and AIC = 542.3745 increased. so we will drop the avg_income.

```
##
## Call:
## lm(formula = passenger_demand ~ airport_traffic + fuel_price,
##     data = df_standard)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.69435 -0.56349  0.01128  0.56557  2.21768
##
## Coefficients:
##                                 Estimate            Std. Error t value
## (Intercept)     -0.0000000000000001348  0.0648340823788336740   0.000
## airport_traffic  0.4041770975115771547  0.0655900609577786120   6.162
## fuel_price       0.1399148447759916980  0.0655900609577786120   2.133
##                        Pr(>|t|)
## (Intercept)              1.0000
## airport_traffic 0.00000000397 ***
## fuel_price               0.0341 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9169 on 197 degrees of freedom
## Multiple R-squared:  0.1678, Adjusted R-squared:  0.1593
## F-statistic: 19.85 on 2 and 197 DF,  p-value: 0.00000001395
```

```
## [1] 537.8466
```

adjusted R squared = 0.1593 increased a bit and AIC = 537.8466 dropped.so we will keep fuel_price.

```
##
## Call:
## lm(formula = passenger_demand ~ airport_traffic + fuel_price +
##     avg_ticket_fare, data = df_standard)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.67971 -0.59302 -0.01358  0.59653  2.18607
##
## Coefficients:
##                               Estimate            Std. Error t value
## (Intercept)     -0.0000000000000001139  0.0647948416693481383   0.000
## airport_traffic  0.4100910999844386007  0.0657653859841450666   6.236
## fuel_price       0.1484374523833845716  0.0659961267412847025   2.249
## avg_ticket_fare -0.0729429430399352458  0.0655394720955686932  -1.113
##                       Pr(>|t|)
## (Intercept)             1.0000
## airport_traffic 0.00000000271 ***
## fuel_price              0.0256 *
## avg_ticket_fare         0.2671
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9163 on 196 degrees of freedom
## Multiple R-squared:  0.173,  Adjusted R-squared:  0.1603
## F-statistic: 13.67 on 3 and 196 DF,  p-value: 0.00000003937
```

```
## [1] 538.5866
```

adjusted R squared = 0.1603 increased a bit and AIC = 538.5866 increased a bit.so we will keep avg_ticket_fare.

```
##
## Call:
## lm(formula = passenger_demand ~ airport_traffic + fuel_price +
##     avg_ticket_fare + flight_frequency, data = df_standard)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.54168 -0.56755  0.01749  0.59793  2.25530
##
## Coefficients:
##                                Estimate             Std. Error t value
## (Intercept)      -0.0000000000000008702  0.0647405185267070082 4   0.000
```

```
## airport_traffic   0.41330892978853595032  0.06576950376944873389   6.284
## fuel_price        0.14830344275455645264  0.06594089891423654537   2.249
## avg_ticket_fare  -0.08615007202602102676  0.06647905424927703921  -1.296
## flight_frequency  0.07601491158494300127  0.06593655783354848698   1.153
##                      Pr(>|t|)
## (Intercept)            1.0000
## airport_traffic  0.00000000211 ***
## fuel_price             0.0256 *
## avg_ticket_fare        0.1965
## flight_frequency       0.2504
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9156 on 195 degrees of freedom
## Multiple R-squared:  0.1786, Adjusted R-squared:  0.1617
## F-statistic:  10.6 on 4 and 195 DF,  p-value: 0.00000008611
```

```
## [1] 539.2281
```

adjusted R squared = 0.1617 increased a bit and AIC = 539.2281 increased a bit.so we will keep flight_frequency.

```
##
## Call:
## lm(formula = passenger_demand ~ airport_traffic + fuel_price +
##     avg_ticket_fare + flight_frequency + route_distance, data = df_standard)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.54518 -0.56629  0.01922  0.59768  2.25708
##
## Coefficients:
##                                Estimate              Std. Error t value
## (Intercept)      -0.00000000000000008691  0.06490675814594784598   0.000
## airport_traffic   0.41299445112710969630  0.06624900799352595460   6.234
## fuel_price        0.14806984571359491087  0.06628134637302562915   2.234
## avg_ticket_fare  -0.08615326124856866052  0.06664978973115615046  -1.293
## flight_frequency  0.07636376262870858689  0.06648693491724881610   1.149
## route_distance   -0.00323357810261597732  0.06588758261519807313  -0.049
##                      Pr(>|t|)
## (Intercept)            1.0000
## airport_traffic  0.00000000278 ***
## fuel_price             0.0266 *
## avg_ticket_fare        0.1977
## flight_frequency       0.2522
## route_distance         0.9609
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9179 on 194 degrees of freedom
## Multiple R-squared:  0.1786, Adjusted R-squared:  0.1574
## F-statistic: 8.436 on 5 and 194 DF,  p-value: 0.0000003055
```

```
## [1] 541.2256
```

adjusted R squared = 0.1574 decreased and AIC = 539.2281 increased.so we will drop route_distance.

so our final model is lm5 because it has the highest accuracy, but the AIC is higher than lm3 we will not consider that. because the AIC increased in lm5 is considerably low. so the regression equation would be

passenger_demand = 0.4133(airport_traffic) + 0.1483(fuel_price) - 0.08615(avg_ticket_fare) +0.07601(flight_frequency)

Looking at the regression analysis results, passenger demand in Sri Lanka is clearly driven more by infrastructure limits than by ticket pricing. Overall airport traffic turned out to be the main growth factor, carrying the highest positive coefficient in our model (0.4133, p < 0.001$). This proves that expanding physical capacity. specifically speeding up the BIA Phase II expansion project should be the top priority for aviation authorities like AASL. The second coefficient for fuel price (0.1483) likely reflects broader economic growth periods where high overall travel demand coincided with rising fuel costs, suggesting that state backed fuel hedging programs could help protect local airlines from global oil shocks. Meanwhile, average ticket fares showed a slight negative coefficient (-0.0862), which aligns with standard price elasticity. however, its impact is secondary, meaning airlines can easily handle this using dynamic yield management to attract budget tourists without hurting business travel revenue. Lastly, the positive coefficient for flight frequency (0.0760) highlights the value of schedule convenience, advising slot management teams to boost weekly flight frequencies to key source markets during peak tourism months.

3.Task B

```
## # A tibble: 6 x 4
##   Source                                       Target       Relationship We
##   <chr>                                        <chr>        <chr>        <
## 1 Bandaranaike International Airport (CMB)     Mihin Lanka  Support
## 2 Bandaranaike International Airport (CMB)     Sri Lanka ~  Commercial
```

```
## 3 Bandaranaike International Airport (CMB)       Fuel Suppl~ Regulatory
## 4 Bandaranaike International Airport (CMB)       Customs & ~ Commercial
## 5 Bandaranaike International Airport (CMB)       Cargo Oper~ Operational
## 6 Mattala Rajapaksa International Airport (MRIA) SriLankan ~ Commercial
```

```
## spc_tbl_ [28 x 4] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Source      : chr [1:28] "Bandaranaike International Airport (CMB)" "Band
##  $ Target      : chr [1:28] "Mihin Lanka" "Sri Lanka Air Force" "Fuel Supply
##  $ Relationship: chr [1:28] "Support" "Commercial" "Regulatory" "Commercial"
##  $ Weight      : num [1:28] 7 5 8 5 1 3 2 4 7 4 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   Source = col_character(),
##   ..   Target = col_character(),
##   ..   Relationship = col_character(),
##   ..   Weight = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

1.Creating an un-directed graph

```
## Number of nodes (stakeholders): 15
```

```
##  [1] "Bandaranaike International Airport (CMB)"
##  [2] "Mattala Rajapaksa International Airport (MRIA)"
##  [3] "Civil Aviation Authority of Sri Lanka (CAASL)"
##  [4] "Airport & Aviation Services SL (AASL)"
##  [5] "SriLankan Airlines"
##  [6] "Mihin Lanka"
##  [7] "Sri Lanka Air Force"
##  [8] "Ground Handling Unit"
##  [9] "Air Traffic Control (ATC)"
## [10] "Fuel Supply Companies"
## [11] "Maintenance & Engineering"
## [12] "Customs & Immigration"
## [13] "International Airlines"
## [14] "Tourism Authority"
## [15] "Cargo Operators"
```

```
## Number of edges (relationships): 28
```

```
## Network density: 0.267
```

```
## Is the network fully connected? TRUE
```

```
## Average path length: 2
```

```
## Network diameter: 4
```

```
## Average clustering coefficient: 0.247
```

here in the created unfirected graph as we can see that there are 15 nodes and they are Bandaranaike International Airport (CMB),Mattala Rajapaksa International Airport (MRIA) ,Civil Aviation Authority of Sri Lanka (CAASL),Airport & Aviation Services SL (AASL),SriLankan Airlines,Mihin Lanka Sri Lanka Air Force, Ground Handling Unit,Air Traffic Control (ATC),Fuel Supply Companies, Maintenance & Engineering Customs & Immigration, International Airlines, Tourism Authority Cargo Operators and they are connectd by 28 edges resulting a fully connected network with density 0.267 which suggest that this is a sparse network. here the average path length(distance between two particular nodes) is 2 and the length of the longest geodesic (the longest shortest path between any pair of vertices) is 4.

2.Calculating centrality metrics

```
##
## Cargo Operators
## Customs & Immigration                                                  Cust
## Fuel Supply Companies                                                  Fuel
## Bandaranaike International Airport (CMB)             Bandaranaike Internatio
## Sri Lanka Air Force                                                      Sr
## Mattala Rajapaksa International Airport (MRIA) Mattala Rajapaksa Internation
## Civil Aviation Authority of Sri Lanka (CAASL)   Civil Aviation Authority of 
## Air Traffic Control (ATC)                                          Air Traf
## Maintenance & Engineering                                          Maintena
## Tourism Authority
## Mihin Lanka
## International Airlines                                                Inter
## SriLankan Airlines                                                       S
## Airport & Aviation Services SL (AASL)                     Airport & Aviation S
## Ground Handling Unit                                                    Gro
##                                                Degree Betweenness Closeness
## Cargo Operators                                     7      0.2921    0.6667
## Customs & Immigration                               6      0.1190    0.6087
## Fuel Supply Companies                               5      0.1868    0.5833
## Bandaranaike International Airport (CMB)            5      0.0916    0.5833
## Sri Lanka Air Force                                 5      0.0833    0.5385
## Mattala Rajapaksa International Airport (MRIA)      4      0.0916    0.5185
## Civil Aviation Authority of Sri Lanka (CAASL)       4      0.0751    0.5185
## Air Traffic Control (ATC)                           4      0.0742    0.4828
```

```
## Maintenance & Engineering                              3      0.0531    0.5385
## Tourism Authority                                      3      0.0220    0.4828
## Mihin Lanka                                            3      0.0201    0.4667
## International Airlines                                 3      0.0201    0.4667
## SriLankan Airlines                                     2      0.0247    0.4242
## Airport & Aviation Services SL (AASL)                  1      0.0000    0.4118
## Ground Handling Unit                                   1      0.0000    0.3784
##                                                    Eigenvector
## Cargo Operators                                         0.9832
## Customs & Immigration                                   1.0000
## Fuel Supply Companies                                   0.7905
## Bandaranaike International Airport (CMB)                0.8722
## Sri Lanka Air Force                                     0.7453
## Mattala Rajapaksa International Airport (MRIA)          0.5932
## Civil Aviation Authority of Sri Lanka (CAASL)           0.5076
## Air Traffic Control (ATC)                               0.3504
## Maintenance & Engineering                               0.4643
## Tourism Authority                                       0.4622
## Mihin Lanka                                             0.3864
## International Airlines                                  0.4930
## SriLankan Airlines                                      0.2108
## Airport & Aviation Services SL (AASL)                   0.2196
## Ground Handling Unit                                    0.1766
```

here the degree centrality explains the raw count of direct connections which helps to identify busy hubs. betweenness explains how often a node sits on the shortest path between other pairs, simply it helps to identify bridges. Closeness helps to explain how quickly a node can reach all others. and finally,eigenvecor centrality helps to identify influence by association. here the cargo operators have the highest degree of 7 which is the most connected and busiest node and act as a bridge as well, and also this node has the highest closeness which explains that from cargo operators node we can reach to any other nodes easily. Customs & Immigration is the 2nd most connected node which also act as a between node but not so often as cargo operators. and it also comparatively close to other nodes. then Fuel Supply Companies, Bandaranaike International Airport (CMB),Sri Lanka Air Force has the same degree but only Fuel Supply Companies act as a between node compared to other two. and all 3 have around same closeness.

Air Traffic Control (ATC) is a useful case for interpreting centrality beyond a simple ranking. Its degree centrality is only on mid range, tied with CAASL and MRIA, and its betweenness score, while present, is lower than that of the network's two cut points. This gap between ATC's numerical centrality and its operational role illustrates an important distinction which is a node's importance to a network is not always proportional to how many connections it holds. ATC's significance lies in which stakeholders it sits between airlines, the regulator, and ground operations rather than in the raw number of its ties. This suggests that resilience planning should not treat centrality scores as the sole indicator of operational importance. Some coordinating roles carry criticality that a structural metric alone does not fully capture.

3.Identify the most influential nodes

```
##
## Top 3 by Degree Centrality


## [1] "Cargo Operators"       "Customs & Immigration" "Fuel Supply Companies"


##
## Top 3 by Betweenness Centrality


## [1] "Cargo Operators"       "Fuel Supply Companies" "Customs & Immigration"
```

as we can see here the most connected nodes are the, Cargo Operators ,Customs & Immigration, Fuel Supply Companies, Bandaranaike International Airport (CMB),Sri Lanka Air Force. and the top 5 critical between nodes are Cargo Operators,Fuel Supply Companies,Customs & Immigration, Bandaranaike International Airport (CMB)and Mattala Rajapaksa International Airport (MRIA).

4.Identify critical bridge nodes

```
## + 2/15 vertices, named, from cc37d8a:
## [1] Fuel Supply Companies Cargo Operators


## + 2/28 edges from cc37d8a (vertex names):
## [1] Ground Handling Unit                 --Fuel Supply Companies
## [2] Airport & Aviation Services SL (AASL)--Cargo Operators
```

from the cut nodes and cut edges we can see that Fuel supply companies and Cargo Operators are cut nodes. Becuase Ground handling unit is only connect to to Fuel supply companies, if Fuel Supply Companies fail, Ground Handling Unit becomes completely isolated from the rest of the aviation sector. And also AASL is only connected to Cargo Operators, so if there is a failure on Cargo Operators AASL will be cut off from the whole network.

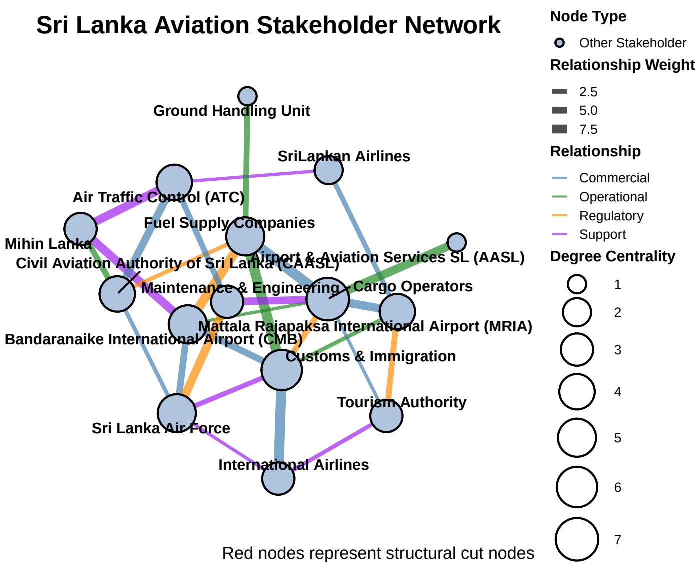
4.Identify clusters of interaction

```
##
## Detected communities (clusters):

##       Bandaranaike International Airport (CMB)
##                                              1
## Mattala Rajapaksa International Airport (MRIA)
##                                              2
##  Civil Aviation Authority of Sri Lanka (CAASL)
##                                              3
##          Airport & Aviation Services SL (AASL)
##                                              2
##                             SriLankan Airlines
##                                              2
##                                    Mihin Lanka
##                                              3
##                             Sri Lanka Air Force
##                                              4
##                            Ground Handling Unit
##                                              1
##                         Air Traffic Control (ATC)
##                                              3
##                           Fuel Supply Companies
##                                              1
##                       Maintenance & Engineering
##                                              4
##                           Customs & Immigration
##                                              1
##                          International Airlines
##                                              1
##                               Tourism Authority
##                                              2
##                                 Cargo Operators
##                                              2

## Modularity score: 0.331
```

as we can see, there are four clusters formed: Cluster 1 : Bandaranaike International Airport (CMB),International Airlines,Fuel Supply Companies Ground Handling Unit,Customs & Immigration and cluster 1 acts as a Ground Operations & Logistics Hub cluster. Cluster 2: Mattala Rajapaksa International Airport (MRIA),SriLankan Airlines,Tourism Authority,Cargo Operators Airport & Aviation Services SL (AASL) and cluster 2 act as a Commercial Aviation & Tourism Aviation cluster. Cluster 3: Sri Lanka Air Force,Maintenance & Engineering, Civil Aviation Authority of Sri Lanka (CAASL) Mihin Lanka. Cluster 4: Maintenance & Engineering

A modularity score of 0.331 indicates moderate community structure. that is stakeholders form meaningfully distinct clusters rather than being uniformly interconnected.

5.Visualizing



3.Task C